Lecture 4

Integrals with parameters and in several variables

Objectives. *An iterated integral needs a hypothesis that permits the change of order. An integral depending on a parameter needs a bound before a limit or a derivative passes inside. We apply these conditions to the Gaussian and then treat an improper integral that is not absolutely integrable.*

By the end of this lecture you will be able to:

- Check nonnegativity or absolute integrability before changing order.
- Use a Jacobian factor in polar coordinates.
- Exhibit a dominating function for a parameter limit or derivative.
- Separate a finite-interval limit from a uniform estimate on a tail.

4.1 Changing the order of integration

The dominated convergence theorem of Theorem 3.5 permits a limit under an integral when a single integrable function bounds the sequence. For two integrations we use the following standing fact. Its cited status has the same inheritance exemption as M1–M3.

Theorem 4.1 (M4: Fubini–Tonelli · *cited*)**.** *Let d_1, d_2 be positive integers. The product-measurability companions are included in M4: a Lebesgue measurable function f on $\mathbb{R}^{d_1}$ lifts to the measurable function $(x, y) \mapsto f(x)$ on $\mathbb{R}^{d_1+d_2}$. In particular, for measurable $A \subset \mathbb{R}^{d_1}$, the product $A \times \mathbb{R}^{d_2}$ is measurable. The same holds for the second coordinate: $(x, y) \mapsto g(y)$ is measurable for measurable g on $\mathbb{R}^{d_2}$. One can also obtain this by interchanging coordinates under M5.*

Let h be Lebesgue measurable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. If $h \geq 0$, its slices are measurable almost everywhere, the iterated integrals are defined as nonnegative integrals, and

$$\int h(x,y)\,\mathrm{d}(x,y) = \int_{\mathbb{R}^{d_1}} \left(\int_{\mathbb{R}^{d_2}} h(x,y)\,\mathrm{d}y \right) \mathrm{d}x = \int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} h(x,y)\,\mathrm{d}x \right) \mathrm{d}y.$$

The values may be infinite. If h is complex valued and $\int |h| < \infty$, the slices are integrable almost everywhere, their integrals are integrable, and the same equalities hold with finite values. Undefined slices may be assigned zero on the exceptional null sets.

This is Stein–Shakarchi, *Real Analysis*, Chapter 2, §3.1, Theorems 3.1–3.2 and Corollary 3.7. Axler, §5B, gives the product-measure form. The almost-everywhere qualification on slices is needed for completed Lebesgue measure. A measurable set of measure zero in a plane can have one nonmeasurable slice.

For example, $h(x,y) = e^{-x^2}e^{-y^2}$ is nonnegative. Tonelli separates its plane integral into a product of two equal one-dimensional integrals. We evaluate it in the next section.

Both iterated integrals can exist without agreeing. On $(0,1)^2$ set

$$h(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}.$$

For fixed $x > 0$, an antiderivative in y is $y/(x^2+y^2)$. For fixed $y > 0$, an antiderivative in x is $-x/(x^2+y^2)$. Consequently,

$$\int_0^1 \left(\int_0^1 h(x,y)\,dy \right) dx = \frac{\pi}{4}, \qquad \int_0^1 \left(\int_0^1 h(x,y)\,dx \right) dy = -\frac{\pi}{4}.$$

The absolute integral is infinite, as the polar-coordinate computation in § 4.2 below will show. Fubini's integrability hypothesis fails.

4.2 Polar coordinates and the Gaussian

Theorem 4.2 (M5: change of variables · *cited*)**.** *Let V and U be open subsets of $\mathbb{R}^d$, and let $\Phi : V \to U$ be a C^1 bijection with C^1 inverse. For a nonnegative measurable function h on U,*

$$\int_U h(x)\,dx = \int_V h(\Phi(y))\,|\det D\Phi(y)|\,dy.$$

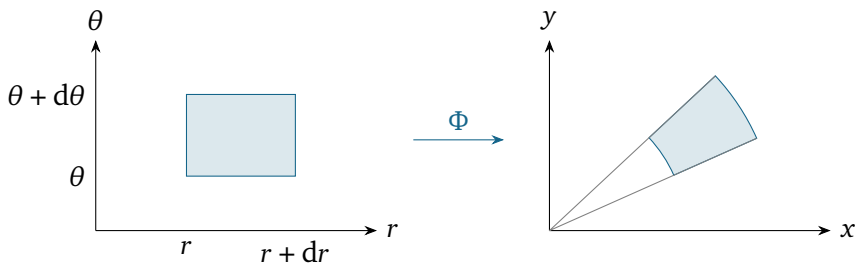
The formula also holds for integrable complex h. M5 also includes the radial formula. For a measurable function h of the radius alone, nonnegative or absolutely integrable with the radial weight, we have

$$\int_{\mathbb{R}^d} h(|x|)\,dx = |S^{d-1}| \int_0^\infty h(r) r^{d-1}\,dr.$$

Here $|S^{d-1}|$ denotes the usual $(d-1)$-dimensional surface area of the unit sphere, not its d-dimensional Lebesgue measure. In dimension one $|S^0| = 2$.

See Strichartz, *The Way of Analysis*, §§15.2.3–15.2.4. The radial formula is interpreted under the same nonnegativity or absolute-integrability condition. Lecture 13 evaluates the sphere area.

For plane polar coordinates, $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$ has determinant r. Take $r > 0$ and $-\pi < \theta < \pi$. This map covers the plane except a ray, which has measure zero. The coordinate cell and its image explain the factor r.



The rectangle has area $dr\,d\theta$ in the coordinate plane. Its image has area factor r, as given by the Jacobian.

For the counterexample in § 4.1, the absolute integrand including its area factor is $|\cos(2\theta)|/r$. On $0 < r < 1$ and $0 < \theta < \pi/8$, the angular factor is at least $1/\sqrt{2}$, while $\int_0^1 \mathrm{d}r/r = \infty$. This proves the asserted failure of absolute integrability.

Theorem 4.3 (Gaussian integral). *The Gaussian satisfies*

$$\int_{\mathbb{R}} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}, \qquad \int_{\mathbb{R}} e^{-ax^2/2}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}} \quad (a > 0). \tag{4.1}$$

Proof. The first integral is finite: outside $[-1, 1]$ its integrand is bounded by $e^{-|x|}$. Write its positive value as J. Tonelli and polar coordinates give

$$J^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)}\,\mathrm{d}(x, y) = 2\pi \int_0^\infty e^{-r^2} r\,\mathrm{d}r = \pi.$$

The positive square root gives $J = \sqrt{\pi}$. The substitution $y = \sqrt{a/2}\,x$ gives the second formula. ∎

4.3 Continuity and differentiation under an integral

An integrable majorant is a nonnegative integrable function that bounds the absolute value of the functions under consideration. The word “single” matters: a bound chosen afresh for each parameter need not justify a limit.

Theorem 4.4 (Parameter integrals). *Let $E \subset \mathbb{R}^d$ be measurable, let $J \subset \mathbb{R}$ be an open interval, and let $f : J \times E \to \mathbb{C}$. Suppose $f(t, \cdot)$ is measurable for each $t \in J$. If $f(t, x)$ is continuous in t for almost every x, and on a neighbourhood of each $t_0 \in J$ it has a single integrable majorant, then $H(t) = \int_E f(t, x)\,\mathrm{d}x$ is continuous.*
For differentiation, suppose instead that $f(t, x)$ is C^1 in t for almost every x, that $f(t_0, \cdot)$ is integrable for one $t_0 \in J$, and that for each compact subinterval $K \subset J$ there are a nonnegative $g_K \in L^1(E)$ and a null set $N_K \subset E$ such that $|\partial_t f(t, x)| \le g_K(x)$ for all $t \in K$ and $x \in E \setminus N_K$. Then H is C^1 and

$$H'(t) = \int_E \partial_t f(t, x)\,\mathrm{d}x. \tag{4.2}$$

Proof. For continuity take any sequence $t_n \to t$ in the neighbourhood. The integrands converge almost everywhere and share its majorant. Dominated convergence gives $H(t_n) \to H(t)$. The sequential criterion for continuity on an interval gives the claim.

For differentiation choose a compact subinterval containing t_0 and t, and a slightly larger one containing $t + h$ for small h. The fundamental theorem of calculus bounds $|f(t, x)|$ by $|f(t_0, x)| + |t - t_0|g(x)$, where g bounds the derivatives there. Thus these functions are integrable. The difference quotients are measurable and satisfy

$$\begin{aligned} &\frac{f(t+h, x) - f(t, x)}{h} \\ &\quad = \int_0^1 \partial_t f(t + uh, x)\,\mathrm{d}u, \\ &\left|\frac{f(t+h, x) - f(t, x)}{h}\right| \le g(x). \end{aligned}$$

They converge almost everywhere to $\partial_t f(t,x)$. Dominated convergence along every sequence $h \to 0$ proves the derivative formula. The derivatives are measurable as limits of measurable difference quotients. Their continuity in t, with the same local majorant, proves continuity of H' by the first part. ∎

For the Gaussian $f(a,x) = e^{-ax^2/2}$, restrict a to $[a_0, a_1]$ with $a_0 > 0$. Its derivative has majorant $(x^2/2)e^{-a_0x^2/2}$, which is integrable since an exponential dominates any polynomial at infinity. Hence

$$\frac{\mathrm{d}}{\mathrm{d}a}\int_{\mathbb{R}} e^{-ax^2/2}\,\mathrm{d}x = -\frac{1}{2}\int_{\mathbb{R}} x^2 e^{-ax^2/2}\,\mathrm{d}x = -\frac{1}{2a}\sqrt{\frac{2\pi}{a}}.$$

The last equality differentiates the value already obtained.

The compact parameter interval keeps a a positive distance from zero. Its minimum a_0 supplies one decay bound for every parameter in that interval. We do not need one majorant for all positive parameters at once. As the interval approaches zero, this particular majorant changes. The parameter theorem proves the local claim stated above.

4.4 The Gaussian transform

Theorem 4.5. *For every $k \in \mathbb{R}$,*

$$\int_{\mathbb{R}} e^{-x^2/2}e^{-\mathrm{i}kx}\,\mathrm{d}x = \sqrt{2\pi}\,e^{-k^2/2}. \tag{4.3}$$

Proof. Call the integral $H(k)$. The integrands and their k derivatives have majorants $e^{-x^2/2}$ and $|x|e^{-x^2/2}$. The parameter theorem gives $H'(k) = -\mathrm{i}\int xe^{-x^2/2}e^{-\mathrm{i}kx}\,\mathrm{d}x$. Since $(e^{-x^2/2})' = -xe^{-x^2/2}$, integration by parts on $[-R,R]$, followed by absolute convergence and the vanishing boundary terms, gives

$$H'(k) = \mathrm{i}\int_{\mathbb{R}} (e^{-x^2/2})' e^{-\mathrm{i}kx}\,\mathrm{d}x = -kH(k).$$

The derivative of $e^{k^2/2}H(k)$ is zero. Its value at zero is $\sqrt{2\pi}$ by the Gaussian integral. This proves the formula. ∎

The Fourier transform in Lecture 7 has the normalising factor $(2\pi)^{-1/2}$, so this Gaussian will equal its own transform.

4.5 An improper integral outside dominated convergence

The function $\sin x/x$, with value 1 at zero, is not integrable on $(0,\infty)$. Indeed, on $[n\pi + \pi/6, n\pi + 5\pi/6]$ its absolute integral is at least $1/(3(n+1))$ for every integer $n \geq 0$. The harmonic series diverges. Its signed improper Riemann integral nevertheless exists.

Theorem 4.6 (Dirichlet integral)**.** *As an improper integral,*

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Proof. For $a > 0$ the absolutely convergent regularisation is

$$I(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} \, \mathrm{d}x.$$

On compact a intervals bounded away from zero, the derivative has integrable majorant $e^{-a_0 x}$. Integration of the elementary complex exponential gives

$$I'(a) = -\int_0^\infty e^{-ax} \sin x \, \mathrm{d}x = -\frac{1}{1+a^2}.$$

Since $|I(a)| \le 1/a$, its limit at infinity is zero. Integrating this derivative gives $I(a) = \pi/2 - \arctan a$.

We must still remove the regularisation. For $a \ge 0$ put $b_a(x) = e^{-ax}/x$. It decreases on $(0, \infty)$. Integration by parts gives, for $T > R > 0$,

$$\left| \int_R^T b_a(x) \sin x \, \mathrm{d}x \right| \le b_a(R) + b_a(T) + \int_R^T |b_a'(x)| \, \mathrm{d}x = 2b_a(R) \le \frac{2}{R}.$$

For $a = 0$ this proves the Cauchy criterion for the improper tail. On $[0, R]$ dominated convergence applies as $a \downarrow 0$, since $|\sin x/x| \le 1$. The difference between the two full integrals is at most the finite-interval difference plus $4/R$. First let $a \downarrow 0$, then let $R \to \infty$. The improper integral equals $\lim_{a \downarrow 0} I(a) = \pi/2$. ∎

The uniform tail estimate replaces the missing integrable majorant on the whole half-line. M3 does not assert agreement for this conditional integral.

Summary.

- Tonelli permits nonnegative integrands. Fubini permits integrable complex integrands. Both include almost-everywhere slice qualifications.
- A coordinate substitution uses the absolute Jacobian determinant. Polar coordinates give the Gaussian integral.
- A local integrable majorant justifies continuity. With integrability at one parameter, an integrable majorant on compact subintervals for the parameter derivative justifies differentiation.
- The Gaussian transform follows from its differential equation.
- The Dirichlet integral is conditional. Its uniform tail estimate is needed when its regularisation is removed.

Lecture 5 uses these integral rules to construct complete function spaces.

References

E. M. Stein and R. Shakarchi, *Real Analysis*, 2005, Chapter 2, §3. R. S. Strichartz, *The Way of Analysis*, revised edition, 2000, §§10.1.4 and 15.2.3–15.2.4. S. Axler, *Measure, Integration & Real Analysis*, 2020, §5B.